



Informatics Institute of Technology

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MATERIALS INCLUDED	REPORT

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Critical and Historical Analysis of an AI Application

1. Historical Context of the Implemented AI Techniques

The AI Travel Assistant developed in Coursework 1 (CW1) combines several Artificial Intelligence techniques instead of relying on a single approach. The main technologies used are **Graph Search**, **Knowledge Representation (KR)**, and **Machine Learning (ML)**. Each of these techniques has developed over many years and plays an important role in the system.

The routing system is based on **Graph Theory** and **Heuristic Search**. In 1959, **Edsger W. Dijkstra** introduced **Dijkstra's Algorithm**, which finds the shortest path between two locations. Although it guarantees the best route, it explores many unnecessary nodes, making it slower for large road networks.

To improve this, **Peter Hart**, **Nils Nilsson**, and **Bertram Raphael** developed the *A Search Algorithm** in 1968. A* uses a heuristic to estimate the remaining distance to the destination, allowing it to search more efficiently while still finding the shortest path. Today, A* is widely used in navigation systems because it is both accurate and fast.

The project also uses **Knowledge Representation (KR)** to make routing decisions based on real world conditions. KR became popular during the 1970s with the development of rule based **Expert Systems**. These systems used simple **IF THEN** rules to solve problems in areas such as medicine and scientific research. In this project, the same idea is used to adjust routes based on conditions such as weather, traffic, and time of day.

The recommendation system is built using **Machine Learning**, specifically a **Decision Tree Regressor**. Decision Trees became widely used after the **Classification and Regression Trees (CART)** method was introduced in 1984 by **Leo Breiman**, **Jerome Friedman**, **Richard Olshen**, and **Charles Stone**. Unlike rule based systems, Decision Trees learn patterns directly from data, making them suitable for predicting different user preferences.

By combining these three AI techniques, the AI Travel Assistant can find efficient routes, adapt to changing conditions, and recommend locations based on different user preferences. This demonstrates how established AI methods can work together to build a more intelligent navigation system.

2. Conceptual Foundations and Paradigms

The AI Travel Assistant developed in CW1 combines different approaches to Artificial Intelligence. Each approach is used for a different purpose within the system.

The routing and context aware features are based on **Rule based AI**. The **A* Search** algorithm follows clear mathematical rules to find the shortest path, while the **Knowledge Representation (KR)** system uses simple **IF-THEN** rules to adjust routes based on conditions such as weather, traffic and time of day. These methods are reliable because they always follow predefined rules.

The recommendation system uses **Machine Learning**, specifically a **Decision Tree Regressor**. Unlike **Rule based** AI, the model learns patterns from data instead of relying on fixed rules. It analyses factors such as detour distance, route distance, review scores, and category matching to predict which POIs best suit different users.

This allows the system to create different recommendations for **Budget** and **Luxury** travellers. For example, Budget users are more likely to receive nearby locations with shorter detours, while Luxury users may be recommended higher rated places even if they require extra travel.

By Graph search with Machine Learning, the system can find efficient routes, adapt to different travel conditions, and provide personalised recommendations. This approach makes the travel assistant more flexible and useful than relying on a single AI technique.

3. Contemporary Applications and Impact

The AI techniques used in this project are widely used in modern applications. They form the foundation of many navigation and recommendation systems used every day.

For example, **A* Search** is commonly used in navigation applications because it can find the shortest route quickly and efficiently. In this project, the evaluation showed that A* explored **87 nodes**, while Dijkstra's Algorithm explored **1,809 nodes** for the same journey. Both algorithms found the same route, but A* completed the search much more efficiently. This makes it more suitable for large road networks and real time navigation.

The project also uses **Decision Tree** models to provide personalised recommendations. Unlike fixed rules, the models learn different user preferences from data. They can recommend different POIs for Budget and Luxury travellers based on factors such as distance, ratings, and category.

Another advantage of Decision Trees is that they are easy to understand. The model can show which features influenced its decisions, making it easier to explain why a particular recommendation was made. This is important because modern AI systems are expected to be transparent and understandable, especially when they influence user decisions.

By combining efficient route planning with personalised recommendations, the AI Travel Assistant demonstrates how established AI techniques can be applied to solve real world navigation problems.

Ethical, Legal and Social Analysis

1. Ethical Implications

The AI Travel Assistant is designed to help users make better travel decisions. However, because it influences the routes people take, it is important to consider the ethical issues involved.

User Safety

The main goal of the system is to provide useful and personalised routes while keeping users safe. Although the routing algorithm aims to find the best route, incorrect or outdated map data could result in unsuitable recommendations. For example, a road may be closed, flooded, or unsafe, but the system may still recommend it if the information is not updated.

For this reason, user safety should always be given higher priority than finding the shortest or fastest route.

Human Control and Decision Making

People can sometimes trust AI systems too much and follow their recommendations without questioning them. This is known as **automation bias**. To reduce this risk, the system is designed to support users rather than make decisions for them.

In this project, both the baseline route and the personalised route are displayed on interactive maps. This allows users to compare the routes and decide which one they prefer, keeping the final decision under human control.

Transparency and Explainability

Another important ethical consideration is transparency. Users should be able to understand why the system makes a particular recommendation.

This project uses **Decision Tree** models because they are easier to explain than many other Machine Learning models. The model can show which features, such as distance, ratings, or detour, had the greatest influence on its recommendations. This makes the system more transparent and easier to understand.

Fairness

The recommendation system must also treat users fairly. In this project, the Budget and Luxury profiles are designed to represent different travel preferences. However, these profiles are based on predefined assumptions and may not reflect the preferences of every user.

For example, not all Budget travellers only care about distance, and not all Luxury travellers always choose the highest rated places. In a real world application, the system should learn from a wider range of users to provide recommendations that are more balanced and fairer.

Overall, the project aims to demonstrate how AI can improve navigation while highlighting the importance of safety, transparency, fairness, and human decision making when developing intelligent systems.

2. Equity, Diversity and Inclusion (EDI)

An AI system should provide fair and useful recommendations for all users, regardless of their background, income, or physical ability. This project highlights several areas where future improvements are needed.

Bias in Map Data

The system relies on data from **OpenStreetMap (OSM)**, which is created and maintained by volunteers. Some areas are mapped in much greater detail than others. As a result, cities or wealthier areas may contain more accurate road and POI information than rural or less developed locations. This can affect the quality of routes and recommendations provided by the system.

User Diversity

The current system only supports two user profiles: **Budget** and **Luxury**. While these profiles are useful for demonstrating personalised recommendations, they cannot represent the wide range of real user preferences. For example, a user may prefer low cost restaurants but still require highly accessible transport or safer routes.

Future versions of the system should support more flexible user profiles and allow users to customise their own preferences instead of selecting from only two options.

Accessibility

The current system does not consider accessibility requirements when generating routes. Users with disabilities or reduced mobility may need routes that include features such as wheelchair accessible paths, ramps, lifts, or pedestrian crossings.

Adding accessibility information in future versions would make the system more inclusive and allow it to recommend routes that better meet the needs of different users.

Overall, improving data quality, supporting more diverse user preferences, and considering accessibility would help make the AI Travel Assistant fairer and more inclusive for a wider range of users.

3. Sustainability and Environmental Impact

Artificial Intelligence systems require computing resources, and the environmental impact of these resources should be considered during development. This project addresses sustainability by focusing on efficient algorithms and lightweight Machine Learning models.

Algorithm Efficiency

One of the main sustainability benefits of this system comes from using **A* Search** instead of only relying on Dijkstra's Algorithm. In testing, A* explored **87 nodes** compared to Dijkstra's **1,809 nodes** while producing the same optimal route.

Reducing the number of calculations means the system requires less processing power, which can reduce energy usage when running many route searches. For a large scale navigation system handling millions of requests, this improvement could significantly reduce computing costs and environmental impact.

Efficient Machine Learning Models

The project uses a **Decision Tree Regressor** instead of more complex models such as deep neural networks. Decision Trees require less computing power and can be trained quickly on standard hardware while still achieving strong prediction accuracy.

This shows that effective AI solutions do not always require large and energy intensive models. Choosing a suitable model based on the problem requirements can help reduce unnecessary resource consumption.

Potential Rebound Effect

Although efficient AI can reduce energy use, there can also be unintended effects. By making travel planning easier and faster, the system may encourage more people to take additional trips by car instead of choosing public transport, cycling, or walking.

Future improvements should consider sustainable travel options, such as recommending public transport routes, walking paths, or lower emission alternatives where possible. This would help ensure that the system supports both efficient travel and environmental responsibility.

4. Legal and Regulatory Considerations

The AI Travel Assistant involves handling location data, user preferences and external geographic information. Therefore, legal and regulatory requirements must be considered to ensure the system is used responsibly.

Data Protection and Privacy

A real world travel assistant would process sensitive user information, including location history, travel patterns, and personal preferences. This data must be collected with user consent and stored securely. Regulations such as the **General Data Protection Regulation (GDPR)** require organisations to protect personal data and allow users to request deletion of their stored information.

Automated Decision Making

The system uses user profiles such as **Budget** and **Luxury** to personalise recommendations. Although these recommendations do not make legal decisions, users should still understand how their data is used and how the system generates suggestions. Human users should always be able to review and choose whether to follow AI recommendations.

Responsibility and Liability

A possible challenge with AI navigation systems is determining responsibility if incorrect recommendations cause harm. For example, if outdated map information leads a user onto an unsafe road, responsibility could involve the software developer, data providers, or the user. Therefore, AI navigation systems should include clear warnings that recommendations are suggestions and users should always consider real world conditions.

Future Regulations and Responsible AI

Future AI regulations, such as the **EU AI Act**, emphasise important principles including transparency, safety, privacy, and human control. This project supports these principles by using explainable Machine Learning models, automated testing, and allowing users to compare different route options before making decisions.

Open Data and Intellectual Property

The project uses OpenStreetMap data, which is provided under the **Open Database License (ODbL)**. This means the system can use data, but it must follow the requirements of the license, including acknowledging the source and respecting sharing conditions for derived datasets.

Overall, legal compliance, privacy protection, and responsible use are essential for developing a trustworthy AI travel system.

Conclusion

The AI Travel Assistant developed in coursework 1 demonstrates how different AI techniques can work together to create a more intelligent and personalised navigation system. By combining heuristic search algorithms, rule based reasoning and Machine Learning, the system can provide efficient routes while considering user preferences and real world conditions.

But developing an efficient AI system requires more than technical performance but which the importance of considering ethical, legal and social factors such as user privacy, data quality, fairness, transparency and environmental impact.

AI system should be designed responsibly with clear explanations, reliable data and human control over final decisions. By balancing technical innovation with responsible development practices,

AI applications like this prototype can provide useful solutions while supporting safe, fair and trustworthy experience for users.

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Supporting Materials

Refer link for more info: [SadushaNadeesh/Foundations-of-Artificial-Intelligence · GitHub](https://github.com/SadushaNadeesh/Foundations-of-Artificial-Intelligence)